

Chapter 5

1.

Using the following rules:

$$Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$$

$$Var(cX) = c^2 Var(X)$$

$$Cov(cX, Y) = Cov(X, cY) = cCov(X, Y)$$

Minimizing two-asset financial portfolio:

$$\begin{aligned} & Var(\alpha X + (1 - \alpha)Y) \\ &= Var(\alpha X) + Var((1 - \alpha)Y) + 2Cov(\alpha X, (1 - \alpha)Y) \\ &= \alpha^2 Var(X) + (1 - \alpha)^2 Var(Y) + 2\alpha(1 - \alpha)Cov(X, Y) \\ &= \sigma_X^2 \alpha^2 + \sigma_Y^2 (1 - \alpha)^2 + 2\sigma_{XY}(-\alpha^2 + \alpha) \end{aligned}$$

Take the first derivative to find critical points:

$$\begin{aligned} 0 &= \frac{d}{d\alpha} f(\alpha) \\ 0 &= 2\sigma_X^2 \alpha + 2\sigma_Y^2 (1 - \alpha)(-1) + 2\sigma_{XY}(-2\alpha + 1) \\ 0 &= \sigma_X^2 \alpha + \sigma_Y^2 (\alpha - 1) + \sigma_{XY}(-2\alpha + 1) \\ 0 &= (\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY})\alpha - \sigma_Y^2 + \sigma_{XY} \\ \alpha &= \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}} \end{aligned}$$

3.

a

k-fold cross-validation is implemented by taking the set of n observations and randomly splitting into k non-overlapping groups. Each of these groups acts as a validation set and the remainder as a training set. The test error is estimated by averaging the k resulting MSE estimates.

b

i. The validation set approach is conceptually simple and easily implemented as you are simply partitioning the existing training data into two sets. However, there are two drawbacks: (1.) the estimate of the test error rate can be highly variable depending on which observations are included in the training and validation sets. (2.) the validation set error rate may tend to overestimate the test error rate for the model fit on the entire data set.

ii. LOOCV is a special case of k-fold cross-validation with k = n. Thus, LOOCV is the most computationally intense method since the model must be fit n times. Also, LOOCV has higher variance, but lower bias, than k-fold CV.

5.

```
library(ISLR)
summary(Default)
## default      student      balance      income
## No :9667      No :7056      Min.   :    0      Min.   :  772
## Yes: 333      Yes:2944      1st Qu.: 482      1st Qu.:21340
##                               Median : 824      Median :34553
##                               Mean   : 835      Mean   :33517
##                               3rd Qu.:1166      3rd Qu.:43808
##                               Max.   :2654      Max.   :73554
attach(Default)
```

a

```
set.seed(1)
glm.fit = glm(default ~ income + balance, data = Default, family = binomial)
```

b

```
FiveB = function() {
  # i.
  train = sample(dim(Default)[1], dim(Default)[1]/2)
  # ii.
  glm.fit = glm(default ~ income + balance, data = Default, family =
binomial,
    subset = train)
  # iii.
  glm.pred = rep("No", dim(Default)[1]/2)
  glm.probs = predict(glm.fit, Default[-train, ], type = "response")
  glm.pred[glm.probs > 0.5] = "Yes"
  # iv.
  return(mean(glm.pred != Default[-train, ]$default))
}
FiveB()
## [1] 0.0286
```

2.86% test error rate from validation set approach.

c

```
FiveB()
## [1] 0.0236
FiveB()
## [1] 0.028
FiveB()
## [1] 0.0268
```

It seems to average around 2.6% test error rate.

d

```
train = sample(dim(Default)[1], dim(Default)[1]/2)
glm.fit = glm(default ~ income + balance + student, data = Default, family =
binomial,
  subset = train)
glm.pred = rep("No", dim(Default)[1]/2)
glm.probs = predict(glm.fit, Default[-train, ], type = "response")
glm.pred[glm.probs > 0.5] = "Yes"
mean(glm.pred != Default[-train, ]$default)
## [1] 0.0264
```

2.64% test error rate, with student dummy variable. Using the validation set approach, it doesn't appear adding the student dummy variable leads to a reduction in the test error rate.

8.

a

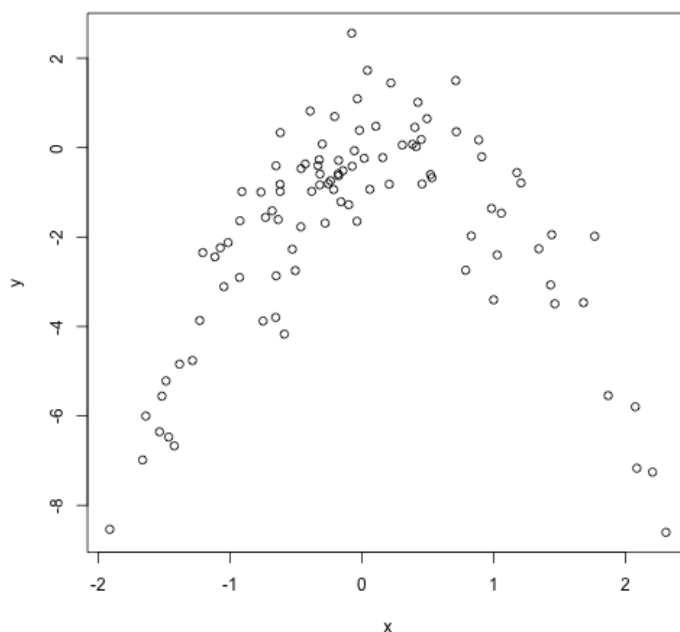
```
set.seed(1)
y = rnorm(100)
x = rnorm(100)
y = x - 2 * x^2 + rnorm(100)
```

$n = 100, p = 2.$

$Y = X - 2X^2 + \epsilon.$

b

```
plot(x, y)
```



Quadratic plot. X from about -2 to 2. Y from about -8 to 2.

c

```
library(boot)
Data = data.frame(x, y)
set.seed(1)
# i.
glm.fit = glm(y ~ x)
cv.glm(Data, glm.fit)$delta
## [1] 5.891 5.889
# ii.
glm.fit = glm(y ~ poly(x, 2))
cv.glm(Data, glm.fit)$delta
## [1] 1.087 1.086
# iii.
glm.fit = glm(y ~ poly(x, 3))
cv.glm(Data, glm.fit)$delta
## [1] 1.103 1.102
# iv.
glm.fit = glm(y ~ poly(x, 4))
cv.glm(Data, glm.fit)$delta
## [1] 1.115 1.114
```

d

```
set.seed(10)
# i.
glm.fit = glm(y ~ x)
cv.glm(Data, glm.fit)$delta
## [1] 5.891 5.889
# ii.
glm.fit = glm(y ~ poly(x, 2))
cv.glm(Data, glm.fit)$delta
## [1] 1.087 1.086
# iii.
glm.fit = glm(y ~ poly(x, 3))
cv.glm(Data, glm.fit)$delta
## [1] 1.103 1.102
# iv.
glm.fit = glm(y ~ poly(x, 4))
cv.glm(Data, glm.fit)$delta
## [1] 1.115 1.114
```

Exact same, because LOOCV will be the same since it evaluates n folds of a single observation.

e

The quadratic polynomial had the lowest LOOCV test error rate. This was expected because it matches the true form of Y.

f

```
summary(glm.fit)
##
## Call:
## glm(formula = y ~ poly(x, 4))
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -2.8913  -0.5244   0.0749   0.5932   2.7796
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.828      0.104  -17.55  <2e-16 ***
## poly(x, 4)1     2.316      1.041    2.22   0.029 *
## poly(x, 4)2  -21.059      1.041  -20.22  <2e-16 ***
## poly(x, 4)3   -0.305      1.041   -0.29   0.770
## poly(x, 4)4   -0.493      1.041   -0.47   0.637
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for gaussian family taken to be 1.085)
##
##      Null deviance: 552.21  on 99  degrees of freedom
## Residual deviance: 103.04  on 95  degrees of freedom
## AIC: 298.8
##
## Number of Fisher Scoring iterations: 2
```

p-values show statistical significance of linear and quadratic terms, which agrees with the CV results.

Chapter 6

- Show that the ridge regression estimate is the maximizer of the (log) posterior distribution, under a Gaussian prior $\beta \sim N(0, \tau^2 I)$, and Gaussian sampling model $y \sim N(X\beta, \sigma^2 I)$. Find the relationship between the regularization parameter λ in the ridge formula, and the variances τ^2 and σ^2 .

The likelihood: $y \sim N(X\beta, \sigma^2 I)$. The prior: $\beta \sim N(0, \tau^2 I)$. Assume that our input data x is already centered (see HTF pg. 59) so that we can ignore the intercept term β_0 . The posterior distribution is:

$$p(\beta|y, X) = \frac{1}{Z} p(y|\beta, X) p(\beta)$$

where $Z = Z(y, X) = \int p(y|\beta, X) p(\beta) d\beta$ is the normalization constant that does not depend on β .

Writing out the (log) posterior distribution explicitly:

$$\begin{aligned} p(\beta|y, X) &= \frac{1}{Z} p(y|\beta, X) p(\beta) \\ &= \frac{1}{Z} \frac{1}{(2\pi)^{n/2} \sigma^n} \exp \left\{ -\frac{1}{2\sigma^2} (y - X\beta)^t (y - X\beta) \right\} \frac{1}{(2\pi)^{p/2} \tau^p} \exp \left\{ -\frac{\beta^t \beta}{2\tau^2} \right\} \\ \log p(\beta|y, X) &= -\log Z - K - \frac{1}{2\sigma^2} (y - X\beta)^t (y - X\beta) - \frac{\beta^t \beta}{2\tau^2} \end{aligned}$$

where K corresponds to the other terms outside the exp. Taking the derivative of the $\log p(\beta|y, X)$ wrt β , we have:

$$\frac{d \log p(\beta|y, X)}{d\beta} = \frac{1}{\sigma^2} X^t (y - X\beta) - \frac{\beta}{\tau^2} \quad (1)$$

Setting this to zero,

$$\begin{aligned} 0 &= \frac{1}{\sigma^2} X^t (y - X\beta) - \frac{\beta}{\tau^2} \\ \rightarrow 0 &= X^t (y - X\beta) - \frac{\sigma^2}{\tau^2} \beta \\ \rightarrow 0 &= X^t y - X^t X \beta - \frac{\sigma^2}{\tau^2} \beta \\ \rightarrow 0 &= X^t y - (X^t X + \frac{\sigma^2}{\tau^2} I) \beta \\ \rightarrow \hat{\beta} &= (X^t X + \frac{\sigma^2}{\tau^2} I)^{-1} X^t y \end{aligned}$$

Setting the ridge regression parameter $\lambda = \frac{\sigma^2}{\tau^2}$, we see that the above solution is equivalent to ridge regression. Note how λ varies with σ^2 and τ^2 . When τ^2 is small (meaning we have prior knowledge to believe β is close to zero), then λ is large (meaning large β will be penalized). When σ^2 is small (meaning that observations y are not noisy), we will focus on fitting the data (small λ).